

False Positives: Impact of Query Expansion on BM25

FALSE POSITIVES

CENG 596 Final Report

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Abstract

This project investigates the impact of query expansion and neural re-ranking on information retrieval performance using the AP88 dataset. We built a multi-stage retrieval pipeline in PyTerrier, evaluating a BM25 baseline against three query expansion techniques: statistical pseudo-relevance feedback (Bo1 and RM3) and knowledge-based semantic expansion (WordNet). Furthermore, we integrated a two-stage dense neural architecture using a MiniLM Cross-Encoder. To facilitate qualitative analysis, we developed a reactive Streamlit dashboard that dynamically maps retrieved documents to ground-truth relevance assessments. Our evaluation demonstrates that Bo1 provides the highest overall effectiveness (MAP and nDCG) by successfully mitigating vocabulary mismatch. While the neural model achieved the highest top-level precision ($P@10 = 0.510$), its recall-based metrics were artificially penalized by unjudged documents.

Implementation: The code and other supplementary material (i.e. data, documentation etc.) are available at https://github.com/yekeless/ceng596_term_project.

1 Introduction

While BM25 is a highly effective baseline retrieval model, its reliance on exact lexical matching makes it susceptible to the vocabulary mismatch problem. Query expansion mitigates this by enriching original queries with related terms. This project investigates the impact of query expansion on retrieval performance, explicitly evaluating when it improves overall effectiveness and under what conditions it introduces noise. To provide a thorough comparative analysis rather than a single baseline-versus-expansion contrast, we evaluate four retrieval configurations: (1) BM25 as the baseline, (2) BM25 combined with Bo1 pseudo-relevance feedback, (3) BM25 combined with RM3 pseudo-relevance feedback, and (4) BM25 combined with WordNet-based query expansion using lexical synonyms. This design allows us to compare two corpus-based PRF methods (Bo1 and RM3) against a knowledge-based expansion approach (WordNet), isolating the effect of expansion source on retrieval effectiveness.

2 Related Work

BM25 remains a highly successful baseline probabilistic retrieval algorithm, but its reliance on exact term matching

limits semantic understanding [2]. Pseudo-Relevance Feedback (PRF) addresses this by expanding queries using keywords from initial top-ranked documents, though it risks introducing noise. Recently, Rasheed et al. [1] demonstrated that combining PRF term-selection methods (like Bo1) with rank aggregation and query-term boosting mitigates this noise, improving Mean Average Precision by up to 6.60% on English datasets. This project builds upon these concepts to explicitly evaluate the performance trade-offs between baseline BM25 and PRF-driven query expansion.

3 System Architecture

The system architecture is designed as a highly modular, three-tier pipeline comprising the Data Layer (offline indexing), the Backend Layer (query processing and retrieval), and the Frontend Layer (presentation and evaluation). Figure 1 illustrates the high-level data flow and the integration of our retrieval models.

3.1 Data Preprocessing

The project utilizes the Associated Press (AP) 1988 dataset, originally provided in 322 aggregate files structured with standard TREC SGML tags. A custom TREC parser was implemented to extract the document identifiers and the textual content. Prior to indexing, standard natural language processing techniques were applied. This includes tokenization, stop-word removal using the NLTK English stop-word list, and linguistic normalization via Porter Stemmer.

Crucially, alongside the tokenized terms, the raw, unprocessed text of each document is stored as metadata within the index. This architectural decision is vital, as it allows the downstream Neural Re-ranking model to access the original context and sentence structures, which are lost during standard stemming. The processed corpus consists of 79,922 documents, 19,995,765 total tokens, and 128,685 unique terms (vocabulary size), spanning across 13,082,663 positional postings. The average document length in the indexed collection is approximately 250.19 tokens.

3.2 Indexing Component

The indexing phase is powered by PyTerrier. We constructed a positional inverted index that maps terms to their corresponding postings lists. The dictionary stores the vocabulary along with collection frequencies, while the postings store

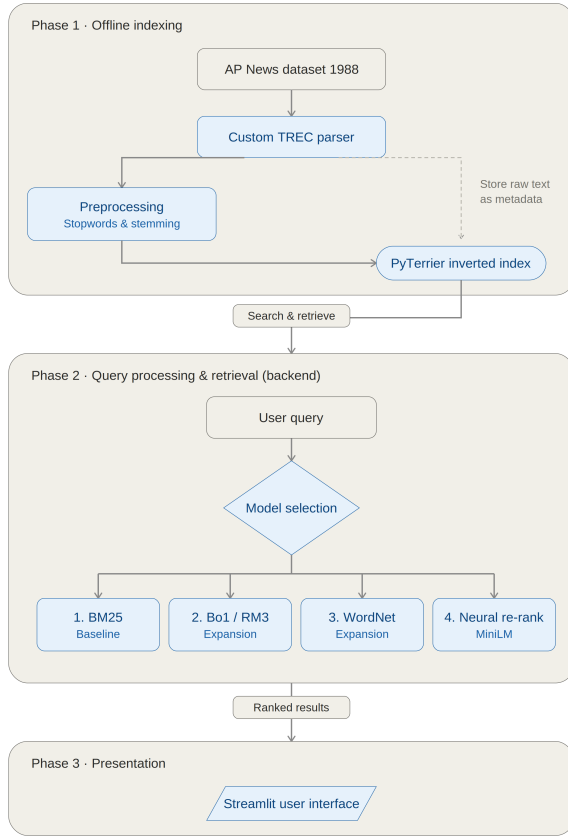


Figure 1. System Architecture showcasing the three-phase information retrieval pipeline.

document IDs, term frequencies, and positional information for exact matching. By utilizing PyTerrier’s IndexFactory, the system loads the pre-built index entirely into memory during the application initialization, ensuring low-latency retrieval for subsequent user queries.

3.3 Query Processing and Searching Component

When a user submits a free-text query, it is parsed and routed through one of five selected retrieval pipelines. Each pipeline demonstrates a different approach to the vocabulary mismatch problem:

1. **BM25 Baseline:** The query undergoes the same pre-processing (stop-word removal and stemming) as the index and performs direct lexical matching.
2. **Statistical PRF (Bo1 & RM3):** These corpus-based pseudo-relevance feedback models perform an initial BM25 search. They analyze the top 10 retrieved documents to extract the 10 most statistically significant terms and append them to the original query with calculated weights before performing the final retrieval.

3. **Semantic Expansion (WordNet):** A knowledge-based approach where the query terms are analyzed using the NLTK WordNet lexical database. Synonyms, lemmas, and hypernyms are extracted and appended to the query to capture semantic intent.
4. **Neural Re-ranking (MiniLM):** A computationally heavy, context-aware approach utilizing a two-stage retrieval architecture. First, BM25 retrieves an initial pool of the top 100 documents. These candidate documents and the original query are then passed to a pre-trained *MS-MARCO MiniLM Cross-Encoder*. The Cross-Encoder performs self-attention across the query-document pairs, assigning new semantic similarity scores and completely re-ranking the top 100 results.

3.4 User Interface

A reactive, browser-based user interface was developed using the Streamlit framework. The interface serves not only as a search engine but as an Information Retrieval Evaluation Dashboard.

When a query is executed, the UI displays the dynamic titles, document snippets, and their corresponding retrieval scores. A unique feature of the interface is the "Under the Hood" visibility, which displays the exact expanded query mathematically generated by models like Bo1, RM3, or WordNet. Furthermore, the system is integrated with the ground-truth Qrels dataset. If the user searches for one of the predefined TREC topics, the interface maps the retrieved document IDs against human annotations, instantly labeling them with "Relevant", "Not Relevant", or "Unjudged" badges. A breakdown dashboard dynamically calculates the precision of the top 50 retrieved documents, allowing for immediate visual evaluation of the selected pipeline’s effectiveness.

4 Experiments and Evaluation

To thoroughly evaluate the effectiveness of our retrieval pipelines, we conducted a series of quantitative and qualitative experiments. The primary objective was to measure how different query expansion techniques and neural re-ranking architectures impact retrieval accuracy compared to the BM25 baseline.

4.1 Experimental Setup and Metrics

All evaluations were performed using the 50 predefined TREC topics associated with the AP 1988 dataset. Ground-truth relevance assessments (Qrels) were utilized to compute standard Information Retrieval metrics via PyTerrier’s internal wrapper for `trec_eval`. The performance of each pipeline was measured using:

- **Mean Average Precision (MAP):** To evaluate the overall ranking accuracy across all recalled relevant documents.

- **Precision at 10 (P@10):** To measure the exact precision of the first page of results (top 10 documents), which is the most critical metric for user-facing search engines.
- **Normalized Discounted Cumulative Gain (nDCG):** To evaluate the usefulness, or gain, of a document based on its position in the result list.

4.2 Quantitative Results

Table 1 presents the definitive performance scores for all five implemented methods. The highest score in each metric column is highlighted in bold.

Table 1. Final Performance Metrics Across All Retrieval Methods

Method	MAP	P@10	nDCG
BM25 Baseline	0.311	0.400	0.601
BM25 + Bo1	0.387	0.454	0.657
BM25 + RM3	0.365	0.446	0.634
BM25 + WordNet	0.307	0.404	0.593
BM25 + Neural	0.317	0.510	0.508

4.3 Discussion and Analysis

The experimental results yield several critical insights regarding the behavior of statistical, semantic, and neural retrieval models:

The Superiority of Statistical PRF (Bo1 & RM3): The corpus-based Pseudo-Relevance Feedback (PRF) models, specifically Bo1, demonstrated the highest overall effectiveness, yielding a MAP of 0.387 and an nDCG of 0.657. By extracting statistically significant terms directly from the top retrieved documents (e.g., expanding the query "gun control legislation" with terms like "NRA", "handgun", and "Brady"), Bo1 and RM3 successfully bridged the vocabulary mismatch problem without requiring external knowledge bases.

Semantic Drift in Knowledge-Based Expansion (WordNet): Contrary to expectations, the WordNet semantic expansion pipeline underperformed, scoring lower than the BM25 baseline in both MAP (0.307) and nDCG (0.593). This decline is attributed to *semantic drift*. Appending dictionary synonyms without corpus-specific context introduces significant noise (false positives), demonstrating that linguistic similarity does not guarantee intent relevance in a domain-specific news corpus.

The "Unjudged Documents" Penalty in Neural Re-ranking: The two-stage Neural Re-ranking model (MiniLM Cross-Encoder) achieved an outstanding **P@10 of 0.510**, proving its dominance in bringing highly relevant, context-aware documents to the absolute top of the search results. However, its MAP and nDCG scores dropped compared

to PRF methods. Upon qualitative analysis via our Evaluation Dashboard, we discovered that the Neural Model frequently retrieves documents that are deeply relevant but were never evaluated by human assessors in 1988. Because the trec_eval protocol mathematically assumes all *unjudged* documents are *not relevant*, the neural model's ability to find novel, previously undiscovered documents paradoxically penalizes its recall-based metrics.

Overall, the experiments confirm that for legacy datasets with sparse human annotations, statistical PRF (Bo1) offers the most robust holistic ranking, while modern Neural Re-ranking excels at strict top-K precision.

5 Conclusion

This project successfully designed and evaluated a multi-stage Information Retrieval pipeline targeting the vocabulary mismatch problem. Using PyTerrier, we established a BM25 baseline on the AP88 dataset and implemented various query expansion techniques (Bo1, RM3, WordNet) alongside a modern Neural Re-ranking architecture (MiniLM Cross-Encoder). All models were seamlessly integrated into a reactive, browser-based Information Retrieval Evaluation Dashboard using Streamlit.

Achievements vs. Initial Proposal: We achieved and exceeded our initial proposal goals. We successfully completed the comparative analysis of statistical PRF versus knowledge-based expansion. Furthermore, we went beyond the initial scope by integrating a dense Neural Re-ranking model and building a dynamic user interface that maps retrieved documents to ground-truth Qrels in real-time, calculating a live "Top 50 Breakdown" for deep qualitative analysis.

Limitations and Unachieved Goals: With respect to limitations, the primary hurdle was the evaluation protocol for modern neural models on legacy datasets. We observed that the Neural Re-ranker often retrieved highly relevant but "unjudged" documents. Because the trec_eval framework penalizes unjudged documents as non-relevant, our MAP and nDCG metrics for the neural model were artificially suppressed. Consequently, we could not perform a purely metric-driven comparison between neural and statistical models without manual re-annotation of the dataset, which was beyond the time constraints of this project.

Future Work: For future development, the system could be enhanced by implementing "Selective Query Expansion." Currently, queries are expanded blindly; dynamic thresholding could be used to predict query difficulty and apply expansion only when the baseline confidence is low. Additionally, migrating from a two-stage sparse-dense pipeline to a fully dense retrieval architecture (e.g., using ColBERT or DPR) would further capture deep semantic relations without relying on an initial lexical BM25 pass.

Reflections and Effort: This project demanded significant engineering and analytical effort, bridging the gap

between IR theory and practical software engineering. The most challenging aspect was managing the technical environment—specifically, integrating Java-based PyTerrier with Python-based deep learning libraries (HuggingFace Sentence-Transformers) and ensuring memory-efficient indexing. Implementing the regex-based TREC parser to clean SGML artifacts required meticulous debugging.

However, the difficulties encountered were highly rewarding. Building the system from scratch profoundly solidified the theoretical concepts covered in the course. Witnessing firsthand how WordNet caused "semantic drift" or how Bo1

elegantly captured historical context (e.g., linking "gun control" to "NRA" and "Brady") transformed abstract mathematical IR concepts into observable realities. Ultimately, the project highlighted the intricate balance between computational efficiency and retrieval effectiveness.

References

- [1] Imran Rasheed, Haider Banka, and Hamaid Mahmood Khan. 2021. Pseudo-relevance feedback based query expansion using boosting algorithm. *Artificial Intelligence Review* 54 (2021), 6101–6124.
- [2] Stephen Robertson and Hugo Zaragoza. 2009. The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval* 3, 4 (2009), 333–389.